

Test-I
CS 602/CO434 Image Processing
21 September, 2020

Marks: 20

Time: 30 minutes

1. Fill up the blanks in the following: 1x5=5
 - a) Digitization of intensity values of an image is called _____.
 - b) An intensity transformation function has a filter size of _____.
 - c) Increasing the pixel values before sending an image for display in a CRT monitor is called _____.
 - d) Image to image transformation is known as _____.
 - e) Image _____ is useful for enhancing white or gray detail embedded in dark regions of an image.
2. Very briefly answer the following: 2x5=10
 - a) State the nature of histogram for a high contrast image.
 - b) Provide mathematical and graphical representation for image negative and log transformation.
 - c) What is the storage requirement of a 1024x1024, 16-level grey scale image?
 - d) Define what is an image histogram.
 - e) Write the coordinates of four-neighbours of pixel $f(x, y)$.
3. Consider the following image. Perform the processing stated below. 1x5=5

2	4	0	6
2	5	1	2
4	6	4	2

- a) Find its negative
- b) Enlarge the image by a factor of 2.
- c) Find the histogram of the image.
- d) Apply histogram sliding with a factor of 2.
- e) Convert the image into a binary image with a threshold value of 4.

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